

Assignment 4 - Predicate Logic, Direct and Indirect Proofs

Due: due Nov. 19 at 11.59pm

Submit your assignment by the due date to Gradescope. Recall that the rules of academic integrity for this class allow you to discuss the questions with your peers, but the write up must be your own. You **may not use any other sources** such as the internet or other textbooks to find solutions to these questions. Your answers will be graded both on their final solution as well as the work you show to achieve that solution. Note that a subset of the questions will be graded. By submitting your assignment you agree to abide by these rules.

1. Let our domain be the natural numbers *greater* than 1. Define:

$$\begin{aligned}P(x) &= \text{"}x \text{ is prime"} \\Q(x, y) &= \text{"}x \text{ divides } y\end{aligned}$$

Consider the following statement:

For every x that is not prime, there is some prime y that divides it. "

- Write the statement in predicate logic.
 - Negate your statement from part (a).
 - Write the English translation of your negated statement. Your statement should sound like English not predicate logic in words.
 - Write the following statement using the given predicates: "*There is exactly one prime number that is even.*". You may **not** use the special quantifier $\exists!$ for *exactly one*.
2. Give an example of a domain U and predicates P and Q such that

$$(\forall x \in U, P(x)) \rightarrow (\forall x \in U, Q(x))$$

is true but

$$\forall x \in U, (P(x) \rightarrow Q(x))$$

is false.

3. Consider the definition of the limit of a sequence in calculus. We can say that the limit of a sequence a_n as n goes to infinity equals L and write this as:

$$\lim_{n \rightarrow \infty} a_n = L$$

if and only if the values of a_n become arbitrarily close to L as n gets larger and larger without bound.

How can we express this more formally?

$$\forall \epsilon \in \mathbb{R}^+, \exists N \in \mathbb{Z}, \forall n \in \mathbb{Z}, n > N \rightarrow L - \epsilon < a_n < L + \epsilon$$

Write the negation of the statement.

4. Prove

$$\forall a \in \mathbb{Z}, \forall b \in \mathbb{Z}, 3|(a^2 + b^2) \rightarrow 3|a \wedge 3|b$$

where $x|y$ means x divides y , i.e., there exists $k \in \mathbb{Z}, xk = y$.

- Prove that the number $\log_5 13$ is irrational. Recall that $\log_b a = x$ means that $b^x = a$.
- Prove that if x is an integer greater than 5 such that both $x - 1$ and $x + 1$ are prime, then x is divisible by 6.
- Given sets A, B and C prove or disprove that $(A - B) \cup (B - C) = (A - B)$.
- Suppose that for integers a, b, n with $n \geq 1$ that $a \equiv_n b$. Prove that $a^2 \equiv_n b^2$.